GEBZE TECHNICAL UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

STUDENT INFORMATION SYSTEM

CSE 102 - Programming Project

User Manual & Comprehensive Documentation

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Version 1.4

Contents

1 Introduction **3**
1.1 Purpose . . . 3
1.2 Document Conventions . . . 3

2 System Architecture **4**
2.1 Module Overview . . . 4
2.2 Data Flow . . . 4

3 Getting Started **5**
3.1 Compilation . . . 5
3.2 Launching the Application . . . 5

4 Main Menu Navigation **5**
4.1 Main Menu Interface . . . 5
4.2 Submenu Structure . . . 6

5 Student Management Module **6**
5.1 Overview . . . 6
5.2 Add New Student . . . 7
5.3 Edit Student Information . . . 8
5.4 Delete Student . . . 8
5.5 Search for Student . . . 9
5.6 Display Student Details and Transcript . . . 10
5.7 Calculate and Display Student GPA . . . 10

6 Course Management Module **11**
6.1 Overview . . . 11
6.2 Add New Course . . . 11
6.3 Edit Course Information . . . 12
6.4 Delete Course . . . 12
6.5 Search for Course . . . 13
6.6 Display Course Details . . . 13
6.7 Set Course Capacity . . . 14
6.8 Assign Prerequisites . . . 14
6.9 List All Courses . . . 14

7 Professor Management Module **15**
7.1 Overview . . . 15
7.2 Add New Professor . . . 15
7.3 Edit Professor Information . . . 16
7.4 Delete Professor . . . 16
7.5 Search Professor . . . 17
7.6 Display Professor Details . . . 17
7.7 Assign Course to Professor . . . 17
7.8 List All Professors . . . 18

8 Enrollment Management Module **18**
8.1 Overview . . . 18
8.2 Enroll Student in Course . . . 19
8.3 Drop Student from Course . . . 19
8.4 View Student's Enrollments . . . 20
8.5 View Class Roster . . . 20

9 Grade Management Module **21**
9.1 Overview . . . 21
9.2 Record New Grade . . . 21
9.3 Update Grade . . . 22
9.4 Calculate Course Statistics . . . 22
9.5 Display Grade Distribution . . . 23

10 Reports Module **23**
10.1 Overview . . . 23
10.2 Student Transcript Report . . . 24
10.3 Course Grade Report . . . 24
10.4 Professor Course Load . . . 24
10.5 Grade Statistics . . . 24

11 System Options Module **25**
11.1 Overview . . . 25
11.2 Save All Data . . . 25
11.3 Reload Data from Files . . . 26

12 Curriculum Management Module **26**
12.1 Overview . . . 26
12.2 Define Degree Requirements (Add/Edit) . . . 27
12.3 Track Student Progress Toward Degree . . . 27
12.4 Show Remaining Requirements . . . 28
12.5 Validate Graduation Eligibility . . . 29
12.6 Auto-Initialize All Curricula (Reset) . . . 29

13 Data File Formats **30**

14 Error Messages Reference **30**

15 Troubleshooting Guide **30**
15.1 Startup & Compilation Issues . . . 30
15.2 Data Operation Issues . . . 30
15.3 Logic & Usage Issues . . . 30
15.4 Display Issues . . . 31

16 Appendix **31**
16.1 Keyboard Shortcuts Summary . . . 31
16.2 GPA Calculation Formula . . . 31
16.3 Semester Format Reference . . . 31

1 Introduction

1.1 Purpose

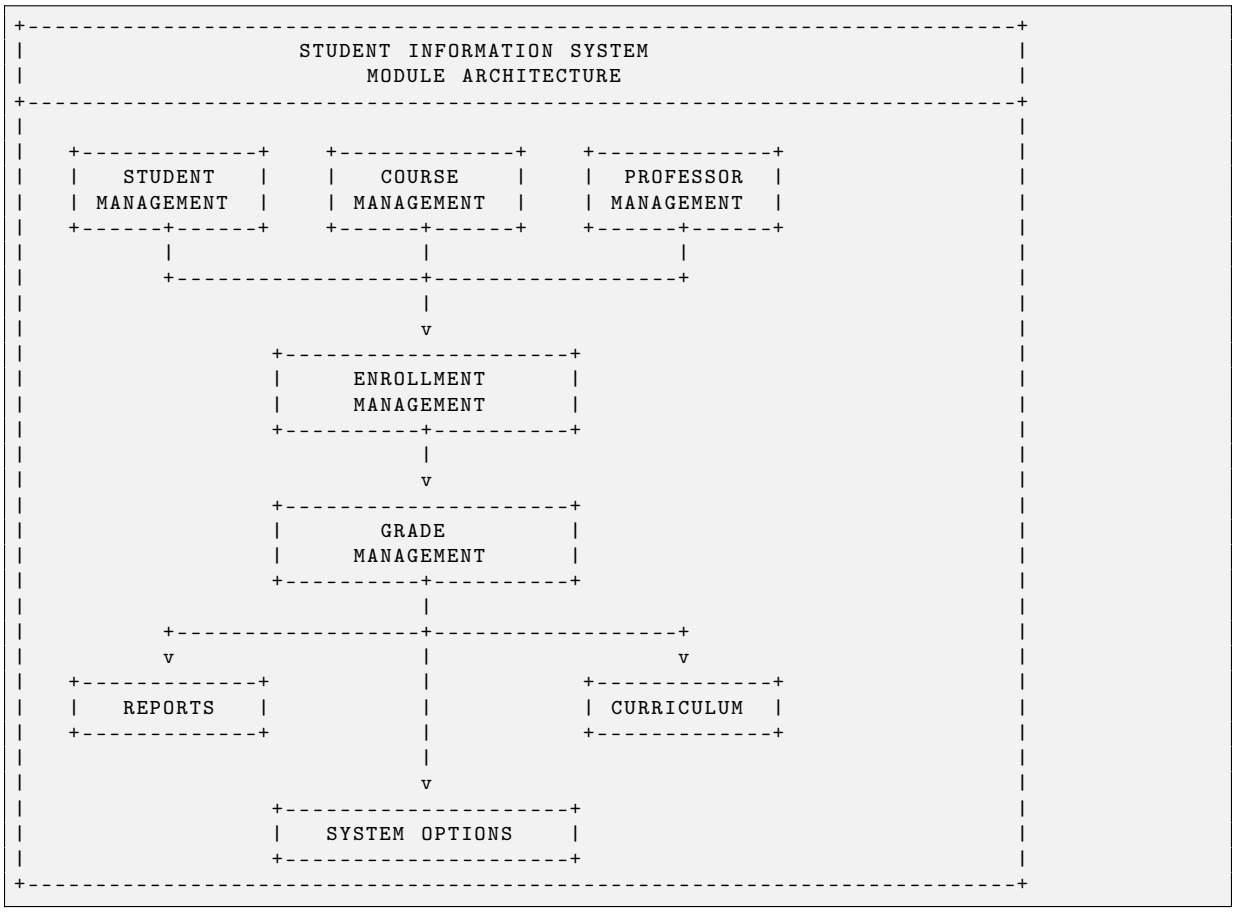
The Student Information System (SIS) is a comprehensive terminal-based application designed to manage academic records for educational institutions.

1.2 Document Conventions

Throughout this manual:

- `Monospace text` indicates commands or menu options
- **Bold text** indicates important concepts
- Screenshot markers indicate where screenshots should be inserted

Figure 1: Student Information System Module Architecture



2 System Architecture

2.1 Module Overview

The system consists of interconnected modules working together to manage student data.

2.2 Data Flow

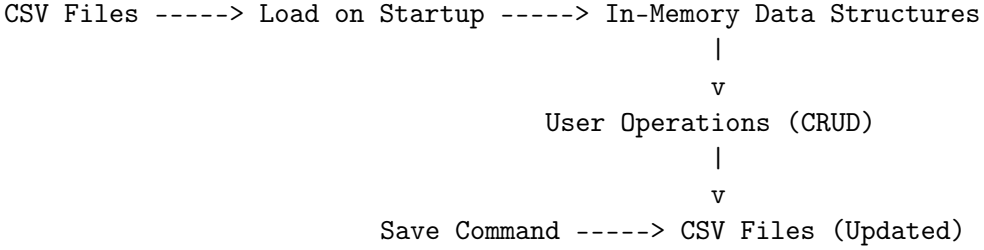
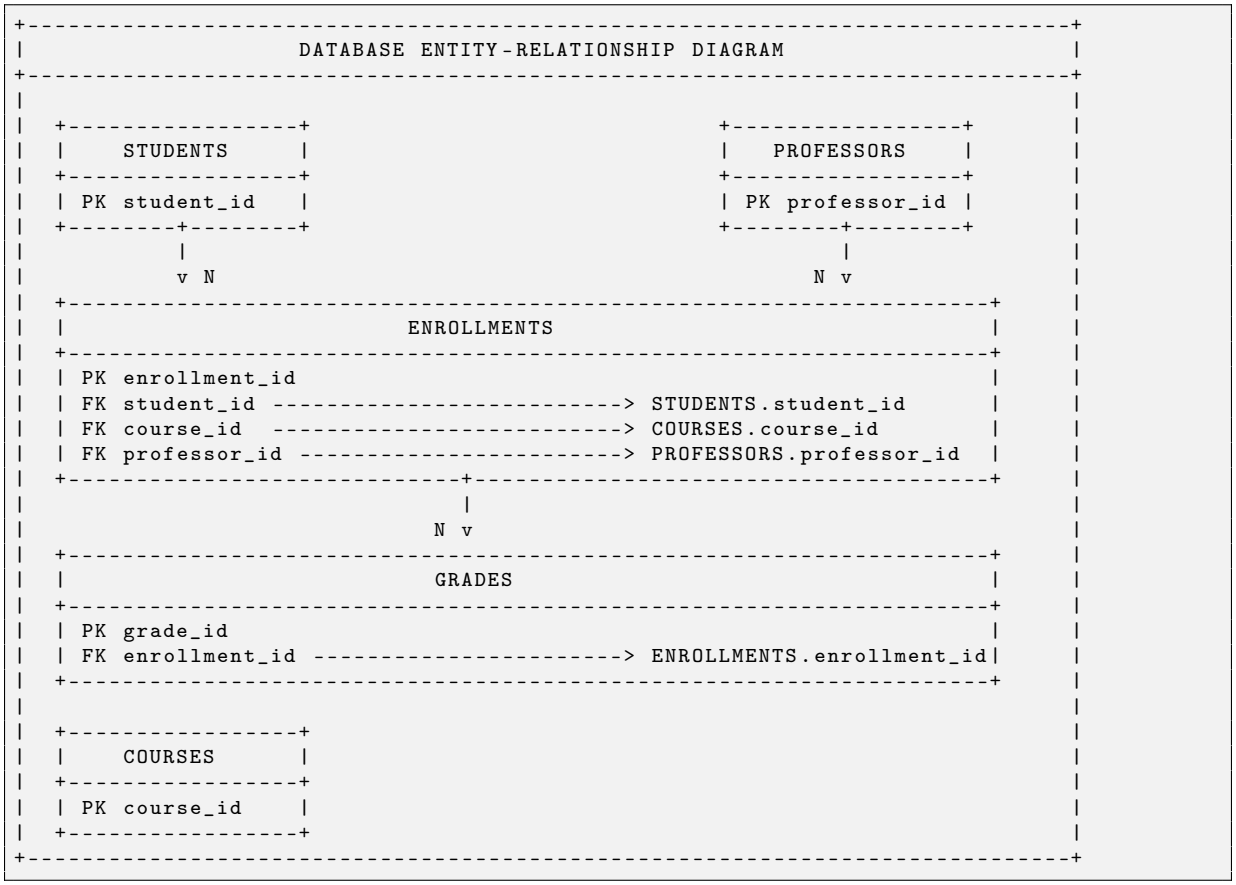


Figure 2: Entity-Relationship Diagram

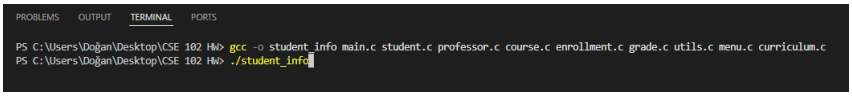


3 Getting Started

3.1 Compilation

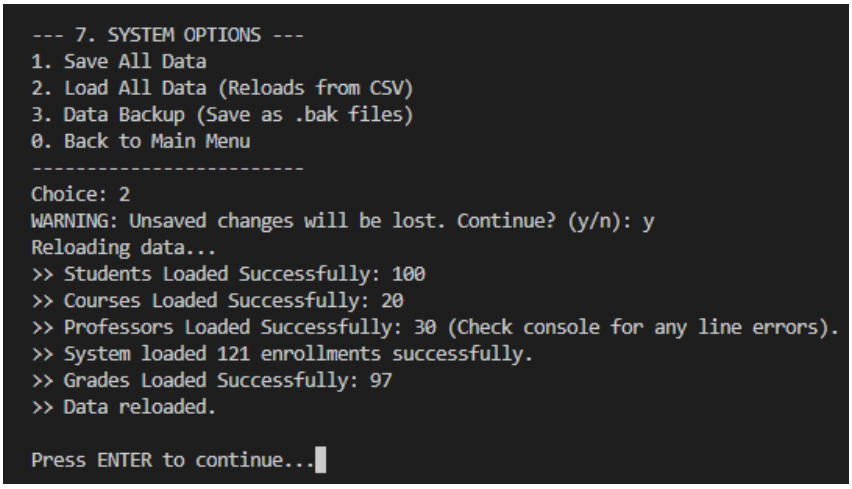
```
make
```

Figure 3.1 - Compilation Terminal Output



3.2 Launching the Application

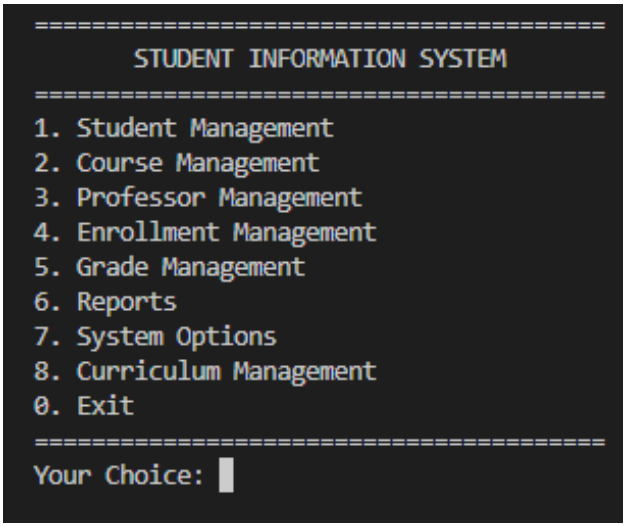
Figure 3.2 - Application Startup



4 Main Menu Navigation

4.1 Main Menu Interface

Figure 4.1 - Main Menu Display



4.2 Submenu Structure

```
Main Menu
    |
    +-- 1. Student Management
    |       (Standard options 1-7)
    |
    +-- 2. Course Management
    |       +-- 1-5. Standard CRUD
    |       +-- 6. Display Course Details
    |       +-- 7. Set Course Capacity
    |       +-- 8. Assign Prerequisites
    |       +-- 9. List All Courses
    |
    +-- 3. Professor Management
    |       +-- 1-4. Standard CRUD
    |       +-- 5. Search Professor
    |       +-- 6. Display Professor Details
    |       +-- 7. Assign Course to Professor
    |       +-- 8. List All Professors
    |
    +-- 4. Enrollment Management
    |       +-- 1. Enroll Student
    |       +-- 2. Update Enrollment
    |       +-- 3. Drop Student from Course
    |       +-- 4. View Student Enrollments
    |       +-- 5. View Class Roster
    |
    +-- 5. Grade Management
    |       +-- 1. Record New Grade
    |       +-- 2. Update Grade
    |       +-- 3. Calculate Course Statistics
    |       +-- 4. Display Grade Distribution
    |
    +-- 6. Reports
    |       +-- 1. Student Transcript
    |       +-- 2. Course Grade Report
    |       +-- 3. Professor Course Load
    |       +-- 4. Grade Statistics
    |
    +-- 7. System Options
    |       +-- 1. Save
    |       +-- 2. Reload
    |       +-- 3. Statistics
    |
    +-- 8. Curriculum Management
            +-- 1. Define Degree Requirements (Add/Edit)
            +-- 2. Track Student Progress Toward Degree
            +-- 3. Show Remaining Requirements
            +-- 4. Validate Graduation Eligibility
            +-- 5. Auto-Initialize All Curricula (Reset)
```

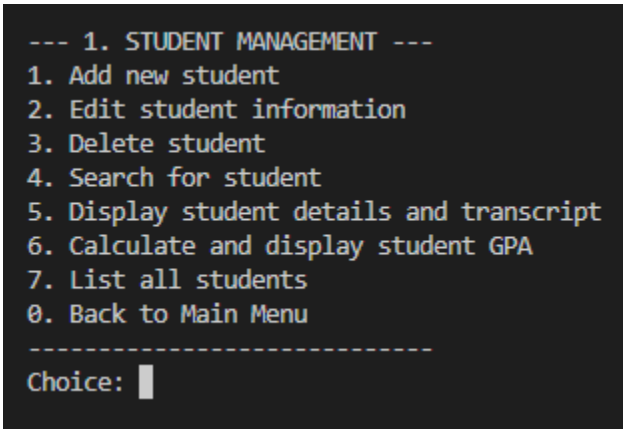
5 Student Management Module

5.1 Overview

The Student Management module is the central hub for maintaining the student database. It allows administrators to register new students with automatic ID generation, update personal information such as emails and majors, and remove records when necessary. Additionally, it provides tools to search for students, view detailed academic transcripts, and calculate GPAs based on course performance.

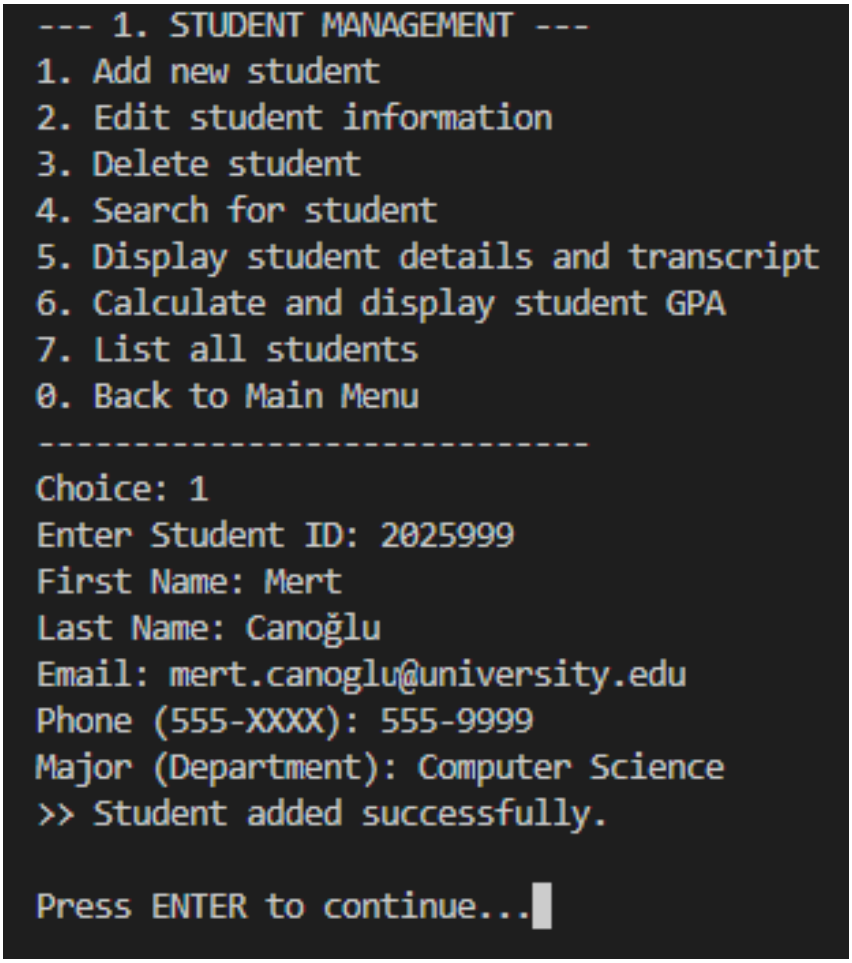
Menu Access: Main Menu → 1

Figure 5.1 - Student Management submenu



5.2 Add New Student

Figure 5.2 - Adding a new student



5.3 Edit Student Information

Figure 5.3 - Editing student info

```
--- 1. STUDENT MANAGEMENT ---
1. Add new student
2. Edit student information
3. Delete student
4. Search for student
5. Display student details and transcript
6. Calculate and display student GPA
7. List all students
0. Back to Main Menu
----------------------------
Choice: 2
Enter Student ID to update: 2025999

--- Update Menu for Mert Canoğlu (2025999) ---
1. Update Name
2. Update Email (Current: mert.canoglu@university.edu)
3. Update Phone (Current: 555-9999)
4. Update Major (Current: Computer Science)
5. Update GPA (Current: 0.00)
0. Back to Main Menu
Choice:
```

5.4 Delete Student

Figure 5.4 - Student deletion

```
--- 1. STUDENT MANAGEMENT ---
1. Add new student
2. Edit student information
3. Delete student
4. Search for student
5. Display student details and transcript
6. Calculate and display student GPA
7. List all students
0. Back to Main Menu
----------------------------
Choice: 3
Enter Student ID to delete: 2025999
WARNING: This will delete the student AND all their enrollments/grades!
Are you sure? (y/n): y
>> Student and all related data deleted successfully.

Press ENTER to continue...
```

5.5 Search for Student

Figure 5.5 - Searching students

```
--- 1. STUDENT MANAGEMENT ---
1. Add new student
2. Edit student information
3. Delete student
4. Search for student
5. Display student details and transcript
6. Calculate and display student GPA
7. List all students
0. Back to Main Menu
----------------------------
Choice: 4
1. ID
2. Name
3. Email
Choice: 1
ID: 2025999

========================================
STUDENT DETAILS
========================================
Student ID      : 2025999
Full Name       : Mert Canoğlu
Email           : mert.canoglu@university.edu
Phone           : 555-9999
Enrollment Year : 2025
Major           : Computer Science
GPA             : 0.00

Press ENTER to continue...
```

5.6 Display Student Details and Transcript

Figure 5.6 - Student transcript

```
========================================
STUDENT DETAILS
========================================
Student ID      : 2020001
Full Name       : Ahmet Yılmaz
Email           : ahmet.yilmaz@university.edu
Phone           : 555-0101
Enrollment Year : 2020
Major           : Computer Science
GPA             : 3.45

==================================================
               OFFICIAL TRANSCRIPT
==================================================
Student ID: 2020001

+-------------------------------------------------------------------------+
| SEMESTER: 2020-FALL                                                     |
+------------+---------------------------------+-------+---------+-------+
| Code       | Course Name                     | Cred  | Grade   | Numeric |
+------------+---------------------------------+-------+---------+-------+
| CS101      | Introduction To Programming     | 4     | B+      | 87.50   |
| MATH101    | Calculus I                      | 4     | A-      | 91.00   |
| MATH202    | Discrete Mathematics            | 3     | B       | 83.50   |
| PHY101     | Physics I                       | 4     | B+      | 88.00   |
+------------+---------------------------------+-------+---------+-------+
| TERM AVG   | Credits: 15                     | 15    | GPA:3.35| 87.50   |
+------------+---------------------------------+-------+---------+-------+
```

5.7 Calculate and Display Student GPA

Figure 5.7 - GPA calculation

```
Choice: 6
Student ID: 2020001

--- GPA Calculation for Ahmet Yılmaz ---
Course ID  Course Name                    Credits    Grade Pt   Weighted
------------------------------------------------------------------------
1001       Introduction To Programming    4          3.30       13.20
2001       Calculus I                     4          3.70       14.80
2004       Discrete Mathematics           3          3.00       9.00
2005       Physics I                      4          3.30       13.20
1002       Data Structures                4          3.70       14.80
2002       Calculus Ii                    4          3.30       13.20
2006       Physics Ii                     4          3.70       14.80
1003       Algorithms                     4          3.70       14.80
2003       Linear Algebra                 3          3.30       9.90
1004       Computer Organization          3          3.30       9.90
1006       Database Systems               4          3.70       14.80
1005       Operating Systems              4          3.30       13.20
1007       Software Engineering           3          3.70       11.10
------------------------------------------------------------------------
Total Credits : 48
Total Points  : 166.70
Calculated GPA: 3.47 / 4.00
>> System Update: Student record updated with new GPA (3.47).
>> (Don't forget to 'Save All Changes' from main menu!)
```

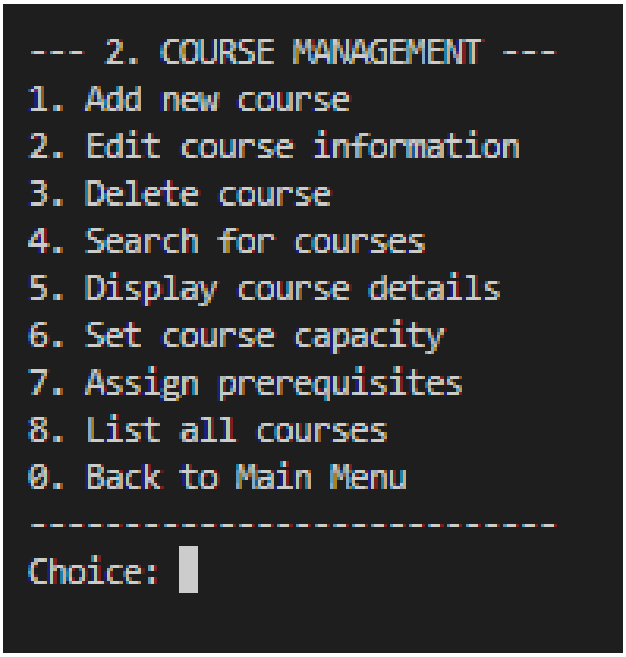
6 Course Management Module

6.1 Overview

This module facilitates the management of the institution's academic catalog. Users can define new courses with specific credit values and department affiliations, set class capacity limits to manage enrollment sizes, and establish prerequisite rules to ensure students take courses in the correct sequence. It also allows for the modification or removal of courses as the curriculum evolves.

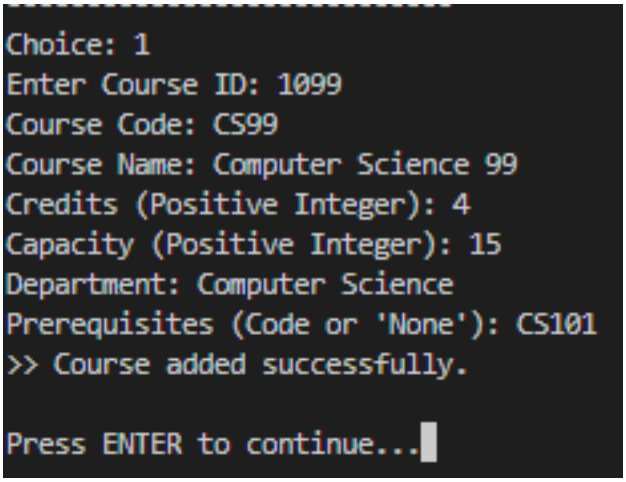
Menu Access: Main Menu → 2

Figure 6.1 - Course Management submenu



6.2 Add New Course

Figure 6.2 - Adding a new course



6.3 Edit Course Information

Figure 6.3 - Modifying course info

```
Choice: 2
Enter Course ID to update: 1099

--- Update Menu for CS99 (1099) ---
1. Update Course Name (Current: Computer Science 99)
2. Update Credits (Current: 4)
3. Update Capacity (Current: 15)
4. Update Department (Current: Computer Science)
5. Update Prerequisites (Current: CS101)
0. Back to Main Menu
Choice: 2
Enter New Credits (>=0): 5

--- Update Menu for CS99 (1099) ---
1. Update Course Name (Current: Computer Science 99)
2. Update Credits (Current: 5)
3. Update Capacity (Current: 15)
4. Update Department (Current: Computer Science)
5. Update Prerequisites (Current: CS101)
0. Back to Main Menu
Choice:
```

6.4 Delete Course

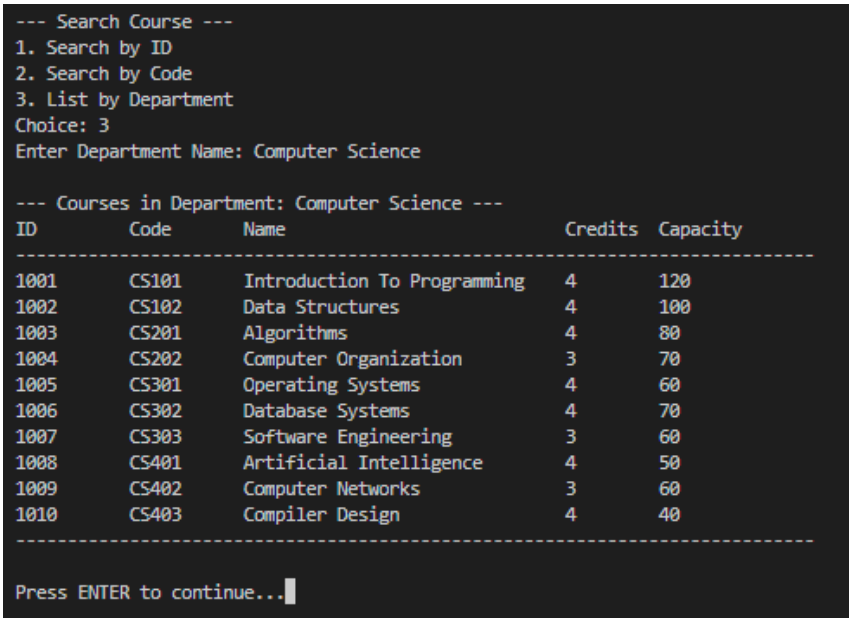
Figure 6.4 - Course deletion warning

```
--- 2. COURSE MANAGEMENT ---
1. Add new course
2. Edit course information
3. Delete course
4. Search for courses
5. Display course details
6. Set course capacity
7. Assign prerequisites
8. List all courses
0. Back to Main Menu
--------------------------
Choice: 3
Enter Course ID to delete: 1099
>> Related grades deleted. New grade count: 97
>> Enrollment records updated. New count: 121
Course deleted successfully.

Press ENTER to continue...
```

6.5 Search for Course

Figure 6.5 - Searching courses

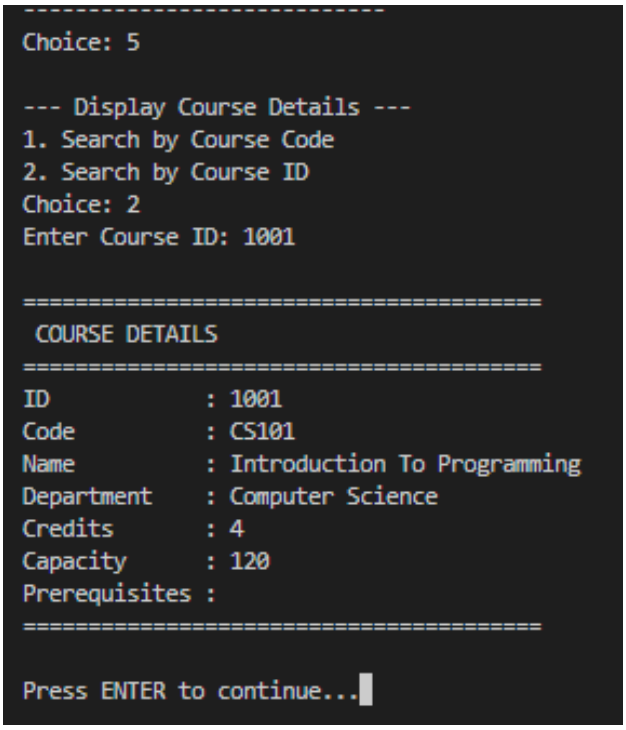


6.6 Display Course Details

Access: Course Management → 6

Displays detailed information about a specific course, including credits, department, and current enrollment count.

Figure 6.6 - Display Course Details

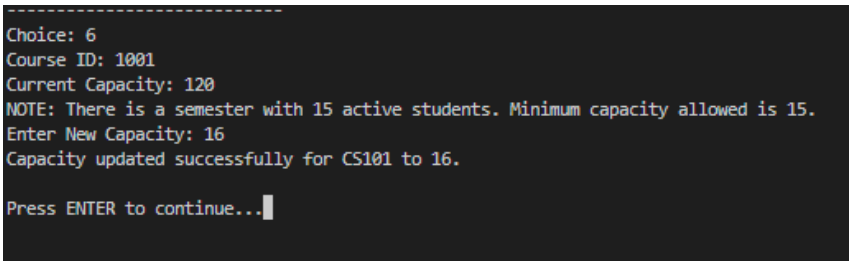


6.7 Set Course Capacity

Access: Course Management → 7

Allows the administrator to set a maximum number of students for a course.

Figure 6.7 - Set Course Capacity

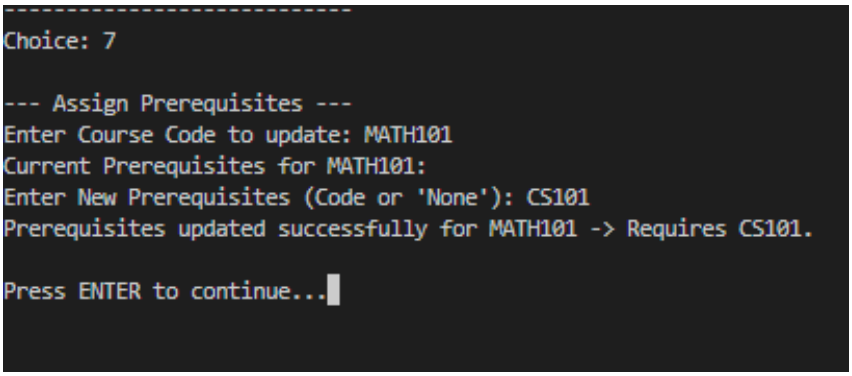


6.8 Assign Prerequisites

Access: Course Management → 8

Define which courses must be completed before enrolling in another course.

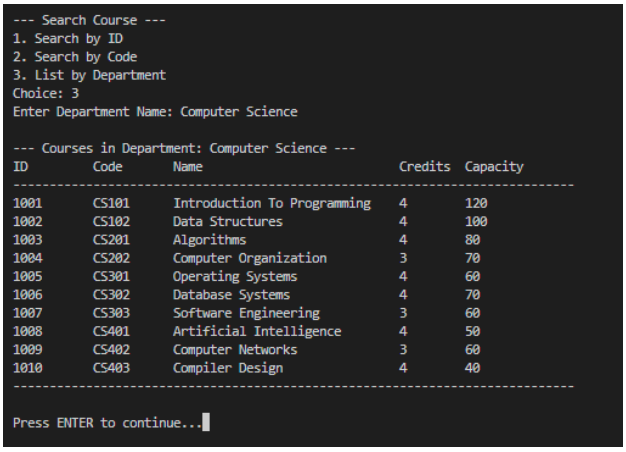
Figure 6.8 - Assign Prerequisites



6.9 List All Courses

Access: Course Management → 9

Figure 6.9 - List All Courses



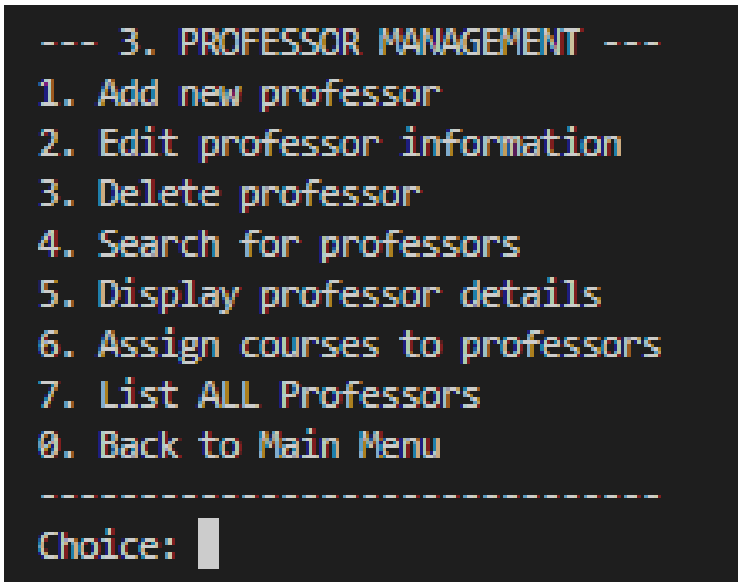
7 Professor Management Module

7.1 Overview

The Professor Management module organizes faculty records. It enables the addition of new faculty members, updates to their contact details and titles, and the management of their departmental assignments. This module is essential for linking instructors to specific course sections and tracking faculty course loads.

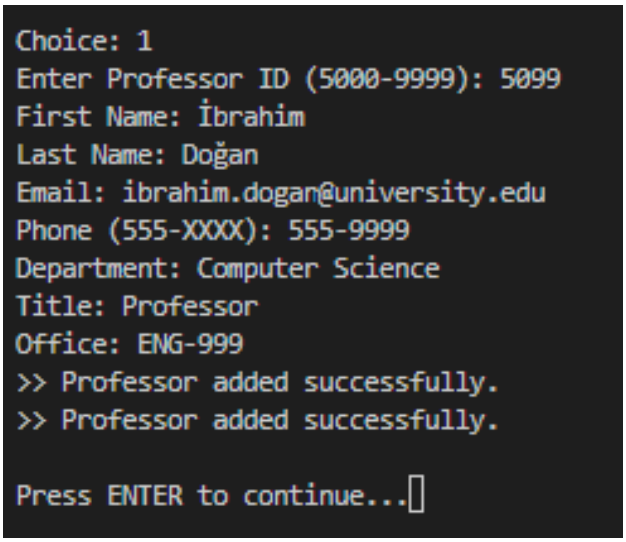
Menu Access: Main Menu → 3

Figure 7.1 - Professor Management submenu



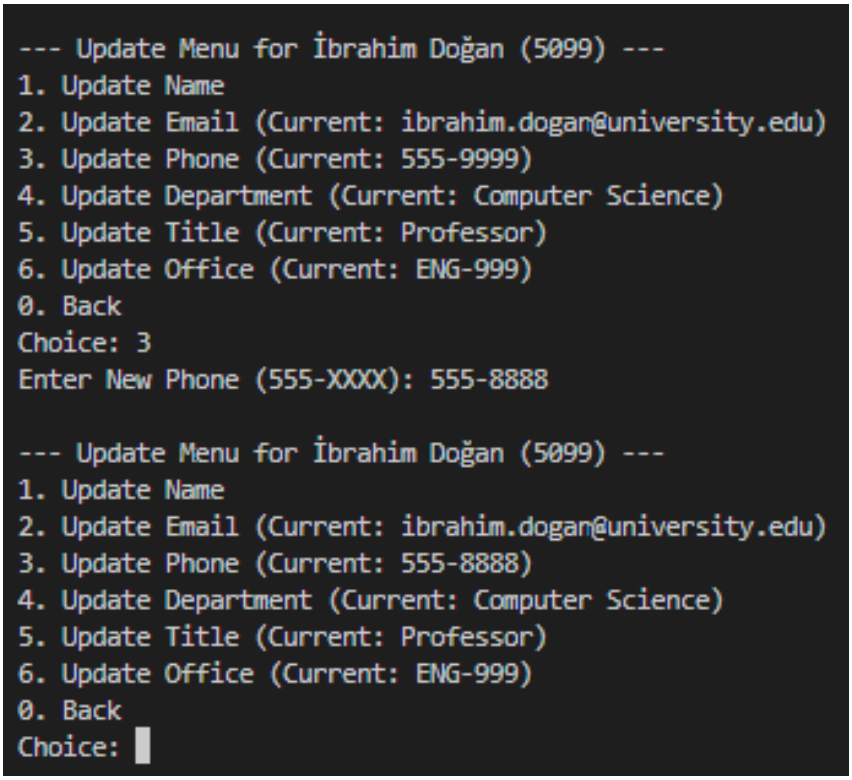
7.2 Add New Professor

Figure 7.2 - Adding a new professor



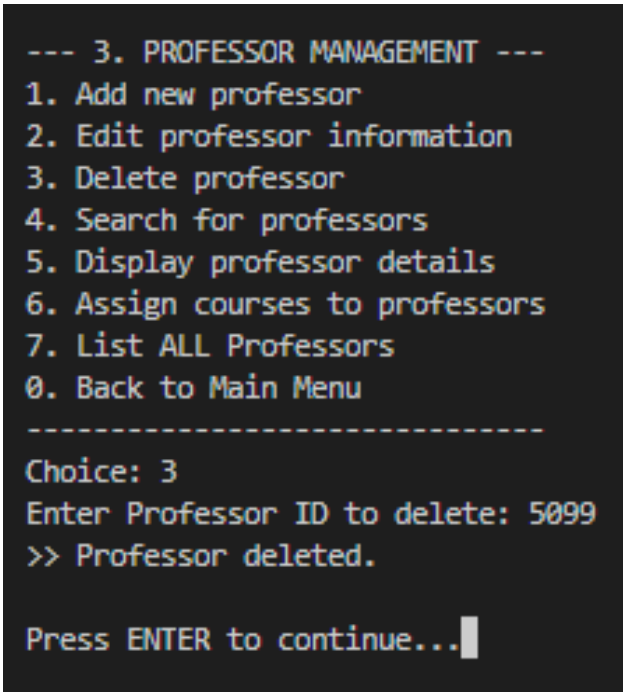
7.3 Edit Professor Information

Figure 7.3 - Updating professor info



7.4 Delete Professor

Figure 7.4 - Professor deletion blocked

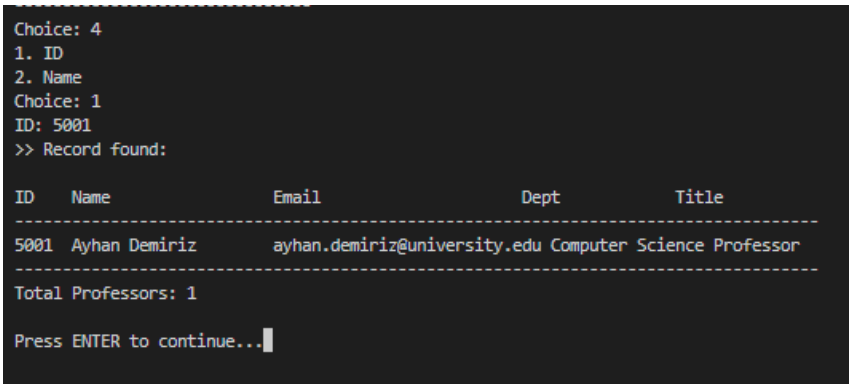


7.5 Search Professor

Access: Professor Management → 5

Search for a professor by name or ID.

Figure 7.5 - Search Professor

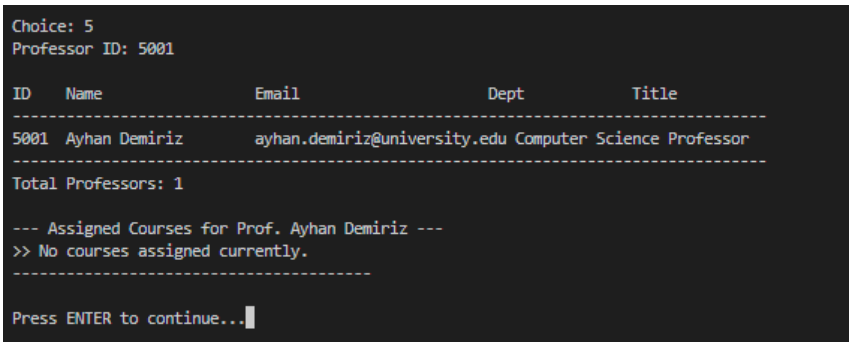


7.6 Display Professor Details

Access: Professor Management → 6

Displays detailed contact info and current course assignments.

Figure 7.6 - Display Professor Details

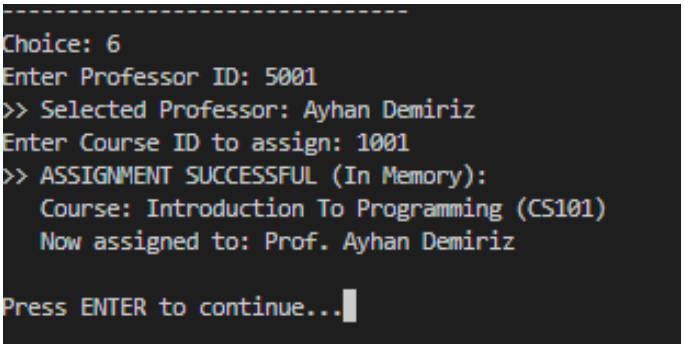


7.7 Assign Course to Professor

Access: Professor Management → 7

Link a course to a professor for a specific semester.

Figure 7.7 - Assign Course to Professor



7.8 List All Professors

Access: Professor Management → 8

Figure 7.8 - List All Professors

```
ID    Name               Email                    Dept          Title
-------------------------------------------------------------------------------
5001  Ayhan Demiriz       ayhan.demiriz@university.edu Computer Science Professor
5002  Serkan Günal       serkan.gunal@university.edu Computer Science Associate Professor
5003  Pınar Karagöz      pinar.karagoz@university.edu Computer Science Professor
5004  Cevat Şener         cevat.sener@university.edu Computer Science Assistant Professor
5005  Nihan Kesim         nihan.kesim@university.edu Computer Science Associate Professor
5006  Cem Say             cem.say@university.edu    Computer Science Professor
5007  Tuğba Taşkaya      tugba.taskaya@university.edu Computer Science Assistant Professor
5008  Selim Aksoy         selim.aksoy@university.edu Computer Science Professor
5009  Yeşim Aydın        yesim.aydin@university.edu Computer Science Associate Professor
5010  Oğuz Altun          oguz.altun@university.edu Computer Science Assistant Professor
5011  Kemal Oflazer       kemal.oflazer@university.edu Computer Science Professor
5012  Gültekin Özsoyoğlu gultekin.ozsoyoglu@university.edu Computer Science Professor
5013  Mustafa Çoşar      mustafa.cosar@university.edu Mathematics     Professor
5014  Ayşe Berkman        ayse.berkman@university.edu Mathematics     Associate Professor
5015  Burak Kanbur         burak.kanbur@university.edu Mathematics     Assistant Professor
5016  Cem Güneri          cem.guneri@university.edu Mathematics     Professor
5017  Zehra İpek          zehra.ipek@university.edu Mathematics     Associate Professor
5018  Ahmet Sezer          ahmet.sezer@university.edu Physics         Professor
5019  Eda Gürel           eda.gurel@university.edu  Physics         Associate Professor
5020  Koray Dağ           koray.dag@university.edu  Physics         Assistant Professor
5021  Levent Onural        levent.onural@university.edu Electrical Engineering Professor
5022  Özgür Gürbüz     ozgur.gurbuz@university.edu Electrical Engineering Associate Professor
5023  Aylin Yener          aylin.yener@university.edu Electrical Engineering Professor
5024  Barış Tan          baris.tan@university.edu  Industrial Engineering Professor
5025  Metin Türkay        metin.turkay@university.edu Industrial Engineering Associate Professor
5026  Tonguç Ünlüyurt   tonguc.unluyurt@university.edu Industrial Engineering Professor
5027  Yaman Arkun          yaman.arkun@university.edu Mechanical Engineering Professor
5028  Melih Papila         melih.papila@university.edu Mechanical Engineering Associate Professor
5029  Mehmet Çalışkan   mehmet.caliskan@university.edu Mechanical Engineering Assistant Professor
5030  Serhat Yeşilyurt    serhat.yesilyurt@university.edu Mechanical Engineering Professor
-------------------------------------------------------------------------------
Total Professors: 30
```

8 Enrollment Management Module

8.1 Overview

This critical module handles the registration of students into academic courses. It enforces business rules such as checking course capacity, verifying prerequisites, and preventing duplicate enrollments. It also allows for dropping students from courses and viewing comprehensive class rosters and individual student enrollment histories.

Menu Access: Main Menu → 4

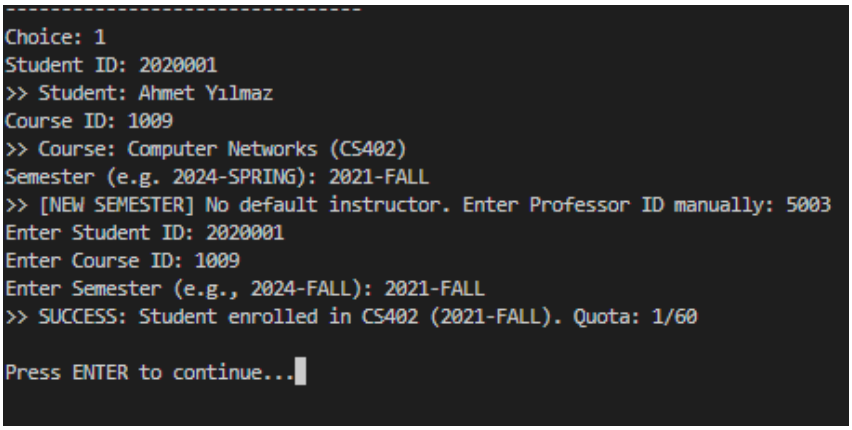
Figure 8.1 - Enrollment Management submenu

```
--- 4. ENROLLMENT MANAGEMENT ---
1. Enroll student in course
2. Drop student from course
3. Display student's current enrollments
4. Display course roster
0. Back to Main Menu
---------------------------------
Choice:
```

8.2 Enroll Student in Course

Access: Enrollment Management → `1`

Figure 8.2 - Enrolling a student

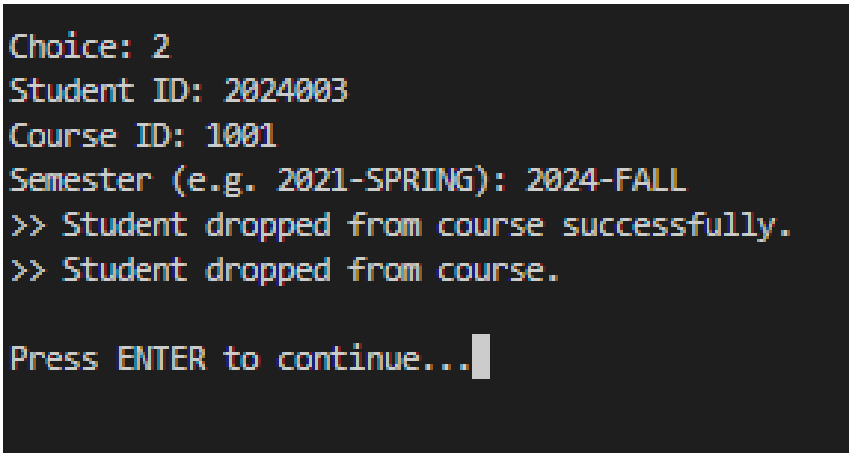


8.3 Drop Student from Course

Access: Enrollment Management → `3`

Removes a student from a course they are currently enrolled in.

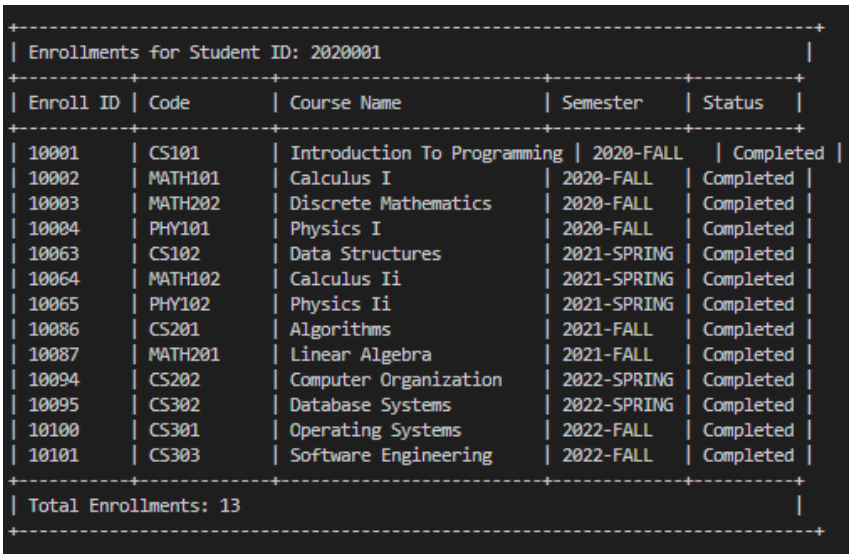
Figure 8.3 - Drop Student from Course



8.4 View Student's Enrollments

Access: Enrollment Management → 4

Figure 8.4 - Student Enrollments

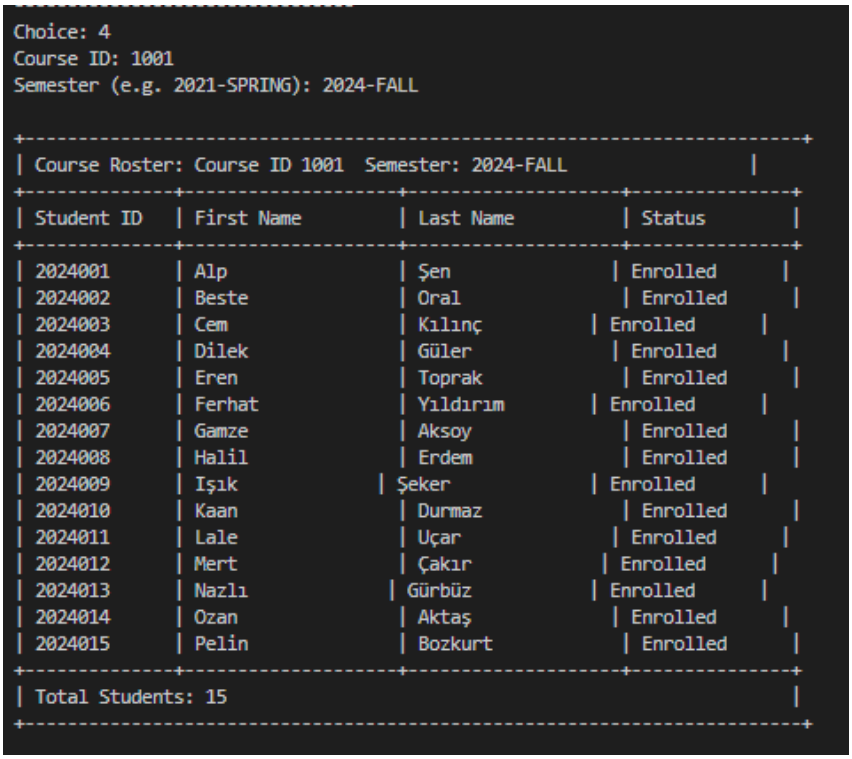


```
+-------------------------------------------------------------------------+
| Enrollments for Student ID: 2020001                                     |
+-----------+------------+-------------------------+-------------+-------+
| Enroll ID | Code       | Course Name             | Semester    | Status |
+-----------+------------+-------------------------+-------------+-------+
| 10001     | CS101      | Introduction To Programming | 2020-FALL   | Completed |
| 10002     | MATH101    | Calculus I              | 2020-FALL   | Completed |
| 10003     | MATH202    | Discrete Mathematics    | 2020-FALL   | Completed |
| 10004     | PHY101     | Physics I               | 2020-FALL   | Completed |
| 10063     | CS102      | Data Structures         | 2021-SPRING | Completed |
| 10064     | MATH102    | Calculus Ii             | 2021-SPRING | Completed |
| 10065     | PHY102     | Physics Ii              | 2021-SPRING | Completed |
| 10086     | CS201      | Algorithms              | 2021-FALL   | Completed |
| 10087     | MATH201    | Linear Algebra          | 2021-FALL   | Completed |
| 10094     | CS202      | Computer Organization   | 2022-SPRING | Completed |
| 10095     | CS302      | Database Systems        | 2022-SPRING | Completed |
| 10100     | CS301      | Operating Systems       | 2022-FALL   | Completed |
| 10101     | CS303      | Software Engineering    | 2022-FALL   | Completed |
+-----------+------------+-------------------------+-------------+-------+
| Total Enrollments: 13                                                   |
+-------------------------------------------------------------------------+
```

8.5 View Class Roster

Access: Enrollment Management → 5

Figure 8.5 - Class Roster



```
Choice: 4
Course ID: 1001
Semester (e.g. 2021-SPRING): 2024-FALL

+-------------------------------------------------------------------------+
| Course Roster: Course ID 1001  Semester: 2024-FALL                      |
+-------------+-------------------+-------------------+-------------+
| Student ID  | First Name        | Last Name         | Status      |
+-------------+-------------------+-------------------+-------------+
| 2024001     | Alp               | Şen               | Enrolled    |
| 2024002     | Beste             | Oral              | Enrolled    |
| 2024003     | Cem               | Kılınç            | Enrolled    |
| 2024004     | Dilek             | Güler             | Enrolled    |
| 2024005     | Eren              | Toprak            | Enrolled    |
| 2024006     | Ferhat            | Yıldırım          | Enrolled    |
| 2024007     | Gamze             | Aksoy             | Enrolled    |
| 2024008     | Halil             | Erdem             | Enrolled    |
| 2024009     | Işık              | Şeker             | Enrolled    |
| 2024010     | Kaan              | Durmaz            | Enrolled    |
| 2024011     | Lale              | Uçar              | Enrolled    |
| 2024012     | Mert              | Çakır             | Enrolled    |
| 2024013     | Nazlı             | Gürbüz            | Enrolled    |
| 2024014     | Ozan              | Aktaş             | Enrolled    |
| 2024015     | Pelin             | Bozkurt           | Enrolled    |
+-------------+-------------------+-------------------+-------------+
| Total Students: 15                                                      |
+-------------------------------------------------------------------------+
```

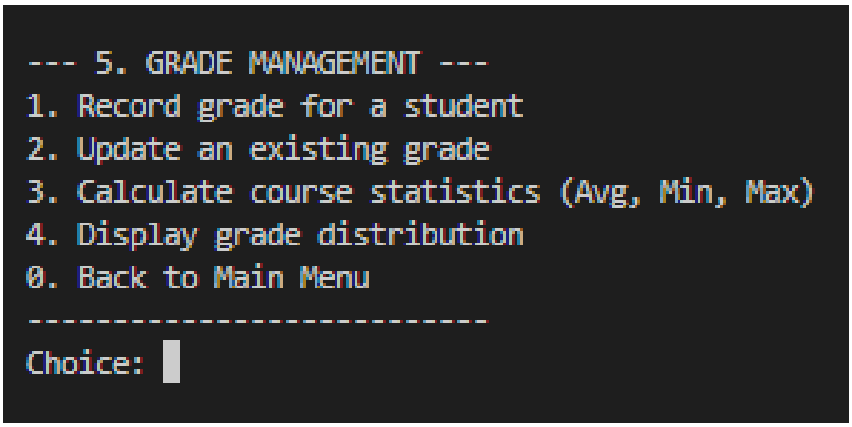
9 Grade Management Module

9.1 Overview

The Grade Management module is responsible for academic assessment. It allows faculty to record numeric grades, which the system automatically converts to letter grades (e.g., A, B, C) based on a standard scale. Users can update existing grades, calculate statistics like class averages, and view visual distributions of student performance for specific courses.

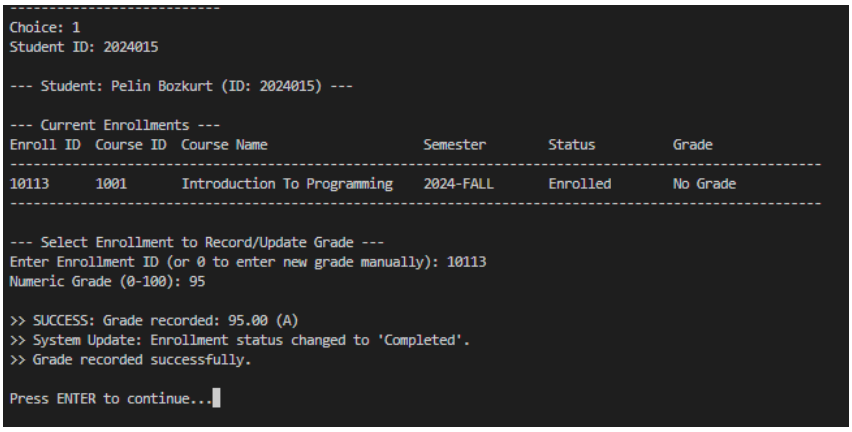
Menu Access: Main Menu → `5`

Figure 9.1 - Grade Management submenu



9.2 Record New Grade

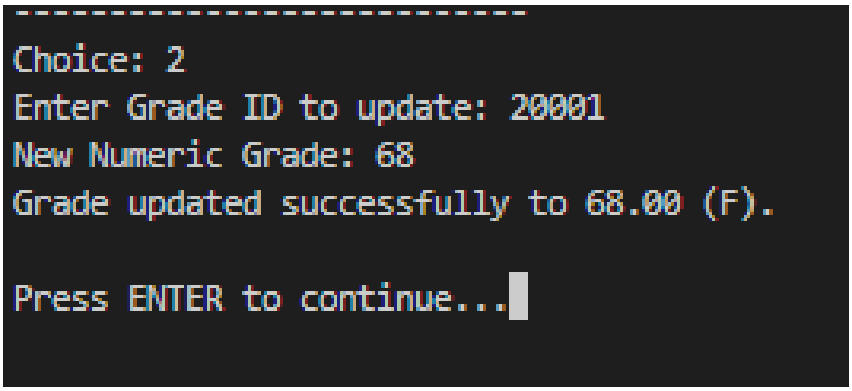
Figure 9.2 - Recording a grade



9.3 Update Grade

Access: Grade Management → 2

Figure 9.3 - Update Grade Process

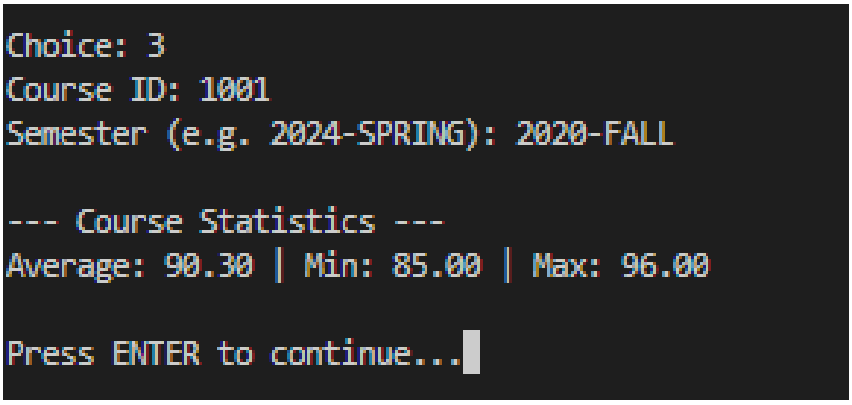


9.4 Calculate Course Statistics

Access: Grade Management → 3

Calculates average, median, and standard deviation for a course's grades.

Figure 9.4 - Calculate Course Statistics

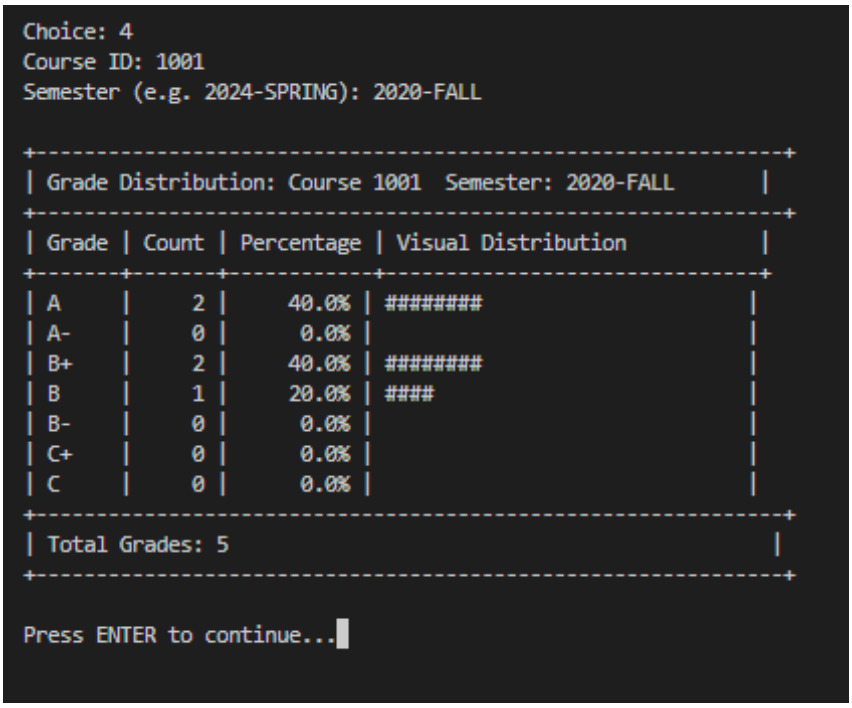


9.5 Display Grade Distribution

Access: Grade Management → 4

Shows a visual histogram of grades (A, B, C, etc.) for the selected course.

Figure 9.5 - Display Grade Distribution



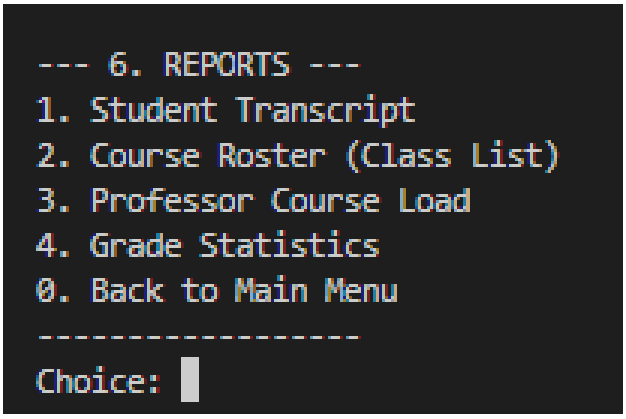
10 Reports Module

10.1 Overview

The Reports module aggregates data from across the system to provide actionable insights. It generates essential documents such as official student transcripts, detailed class rosters for instructors, professor workload reports, and statistical analyses of grade distributions across different courses.

Menu Access: Main Menu → 6

Figure 10.1 - Reports submenu



10.2 Student Transcript Report

Access: Reports → 1

10.3 Course Grade Report

Access: Reports → 2

10.4 Professor Course Load

Access: Reports → 3

Shows how many courses and students each professor is managing.

Figure 10.3 - Professor Course Load

```
------------------
Choice: 3
Enter Professor ID: 5001
Semester (e.g. 2024-SPRING): 2024-FALL

================================================================================
 COURSE LOAD REPORT: Prof. Ayhan Demiriz
 Semester: 2024-FALL
================================================================================
ID    Code        Course Name                  Capacity   Enrolled   Status
--------------------------------------------------------------------------------
1001  CS101       Introduction To Programming  16         2          Open
--------------------------------------------------------------------------------
SUMMARY: 1 Courses | Total Students Handled: 2
================================================================================

Press ENTER to continue...
```

10.5 Grade Statistics

Access: Reports → 4

Detailed statistical analysis of grades across the department.

Figure 10.4 - Grade Statistics

```
--- Grade Statistics ---
Average Grade: 90.40
Min Grade    : 84.50
Max Grade    : 97.00

+-------------------------------------------------------------+
| Grade Distribution: Course 1001  Semester: 2021-FALL        |
+-------------------------------------------------------------+
| Grade | Count | Percentage | Visual Distribution            |
+-------+-------+------------+--------------------------------+
| A     |     2 |     40.0% | ########                       |
| A-    |     1 |     20.0% | ####                           |
| B+    |     1 |     20.0% | ####                           |
| B     |     1 |     20.0% | ####                           |
| B-    |     0 |      0.0% |                                |
| C+    |     0 |      0.0% |                                |
| C     |     0 |      0.0% |                                |
+-------------------------------------------------------------+
| Total Grades: 5                                             |
+-------------------------------------------------------------+

Press ENTER to continue...
```

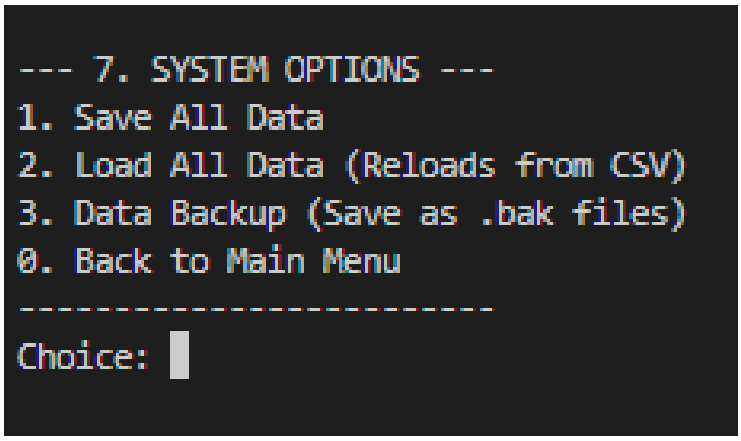
11 System Options Module

11.1 Overview

This administrative module handles the persistence and maintenance of the system's data. It provides functionality to save the current state of the database to CSV files, reload data to revert unsaved changes, and view technical statistics such as memory usage and record counts. It also includes a reset function for initializing a fresh database.

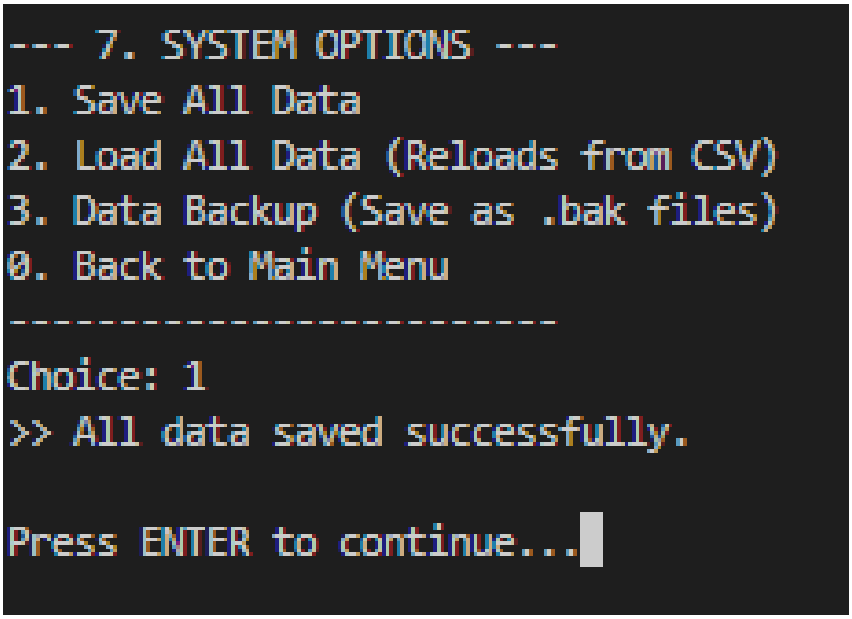
Menu Access: Main Menu → 7

Figure 11.1 - System Options submenu



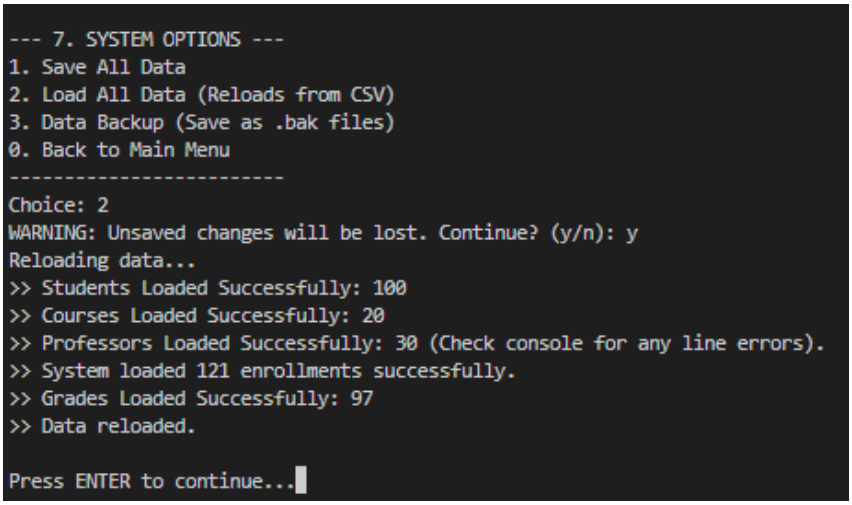
11.2 Save All Data

Figure 11.2 - Saving data to CSV



11.3 Reload Data from Files

Figure 11.3 - Reloading data



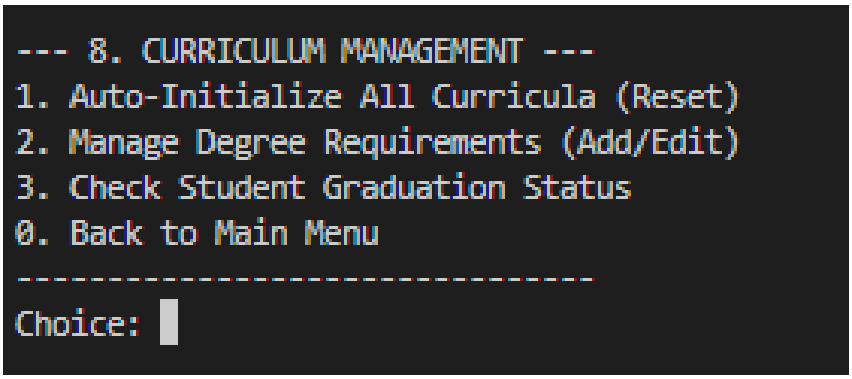
12 Curriculum Management Module

12.1 Overview

The Curriculum Management module defines the academic requirements for graduation. It allows the definition of required courses for each department, tracks individual student progress against these requirements, and validates graduation eligibility. It features an auto-initialization tool that generates default curricula based on existing course data.

Menu Access: Main Menu → `8`

Figure 12.1 - Curriculum Management submenu

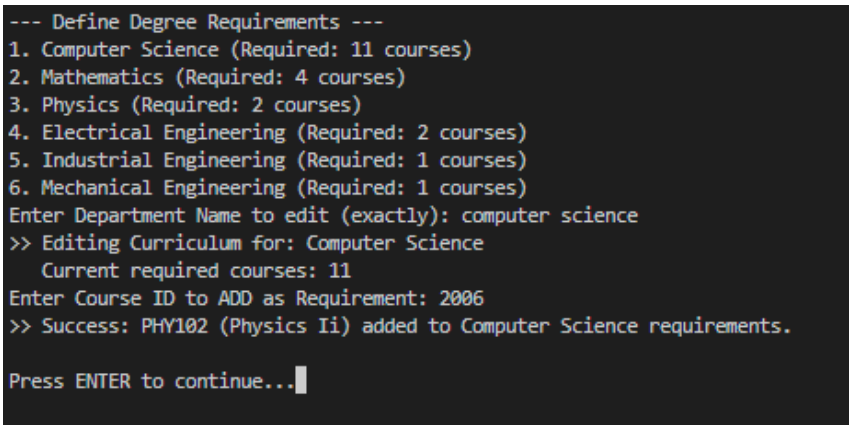


12.2 Define Degree Requirements (Add/Edit)

Access: Curriculum Management → 1

Define required courses and credit totals for specific majors or degree programs.

Figure 12.2 - Define Degree Requirements

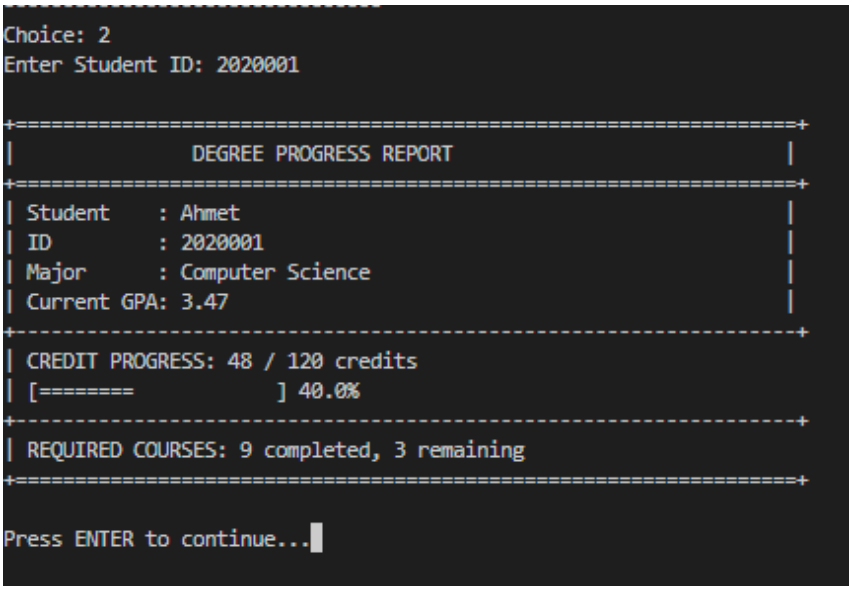


12.3 Track Student Progress Toward Degree

Access: Curriculum Management → 2

View a student's completed courses against the degree requirements.

Figure 12.3 - Track Student Progress

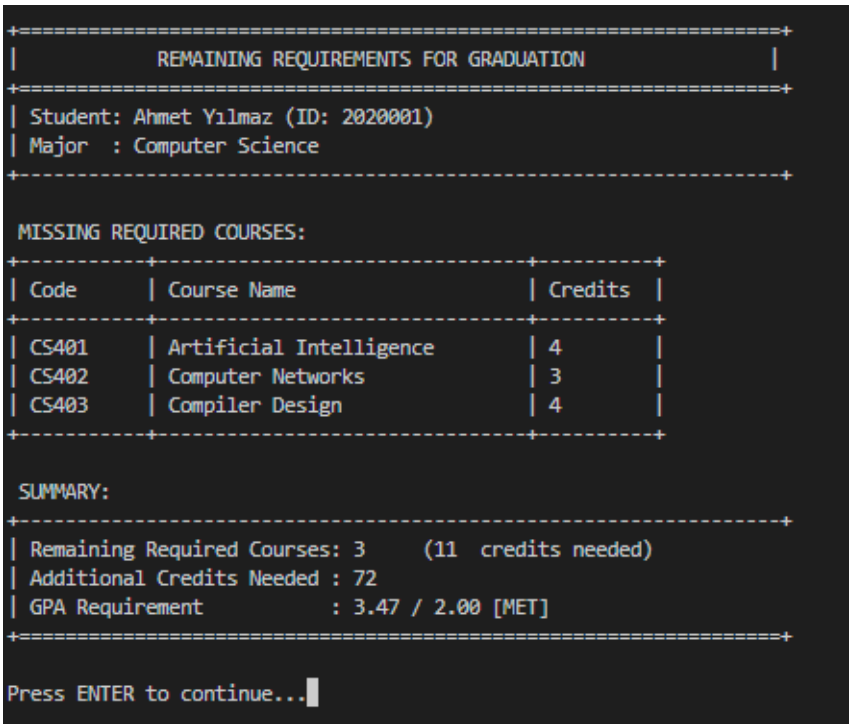


12.4 Show Remaining Requirements

Access: Curriculum Management → `3`

List the courses a student still needs to take to graduate.

Figure 12.4 - Show Remaining Requirements



12.5 Validate Graduation Eligibility

Access: Curriculum Management → 4

Automatically check if a student meets all criteria to graduate.

Figure 12.5 - Validate Graduation Eligibility

```
--------------------------------
Choice: 4
Enter Student ID: 2020001

========================================
        GRADUATION STATUS REPORT
========================================
Student    : Ahmet Yılmaz (2020001)
Department : Computer Science

--- REQUIRED COURSES STATUS ---
Code       Course Name                  Status
--------------------------------------------------
CS101      Introduction To Programming  [COMPLETED]
CS102      Data Structures              [COMPLETED]
CS201      Algorithms                   [COMPLETED]
CS202      Computer Organization        [COMPLETED]
CS301      Operating Systems            [COMPLETED]
CS302      Database Systems             [COMPLETED]
CS303      Software Engineering         [COMPLETED]
CS401      Artificial Intelligence      [MISSING] <--
CS402      Computer Networks            [MISSING] <--
CS403      Compiler Design              [MISSING] <--
PHY101     Physics I                    [COMPLETED]
PHY102     Physics Ii                   [COMPLETED]

--- SUMMARY ---
Credits Earned  : 48 / 120 (Minimum)
CGPA            : 3.47 / 2.00 (Minimum)
Missing Courses : 3
--------------------------------------------------
[-] Failed: Missing required courses.
[-] Failed: Not enough credits (72 needed).
 DECISION: NOT ELIGIBLE.
==================================================

Press ENTER to continue...
```

12.6 Auto-Initialize All Curricula (Reset)

Access: Curriculum Management → 5

Reset the curriculum data to system defaults.

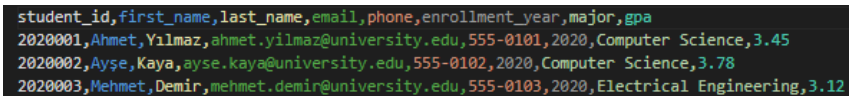
Figure 12.6 - Auto-Initialize Curricula

```
--------------------------------
Choice: 5
Auto-generating default curricula based on course departments...
Initialization complete. 6 departments found.

Press ENTER to continue...
```

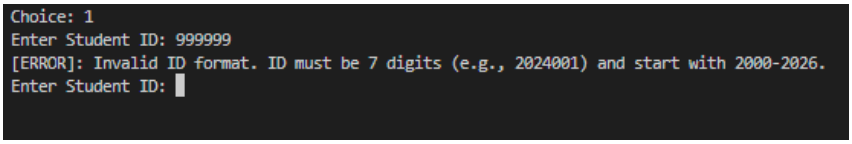
13 Data File Formats

Figure 13.1 - Sample CSV data files



14 Error Messages Reference

Figure 14.1 - Error messages example



15 Troubleshooting Guide

15.1 Startup & Compilation Issues

- **Problem:** "Command not found" or executable does not run.
  - **Cause:** Project not compiled or wrong directory.
  - **Solution:** Run `make`. Use `student_info.exe` (Windows) or `./student_info` (Linux/Mac).
- **Problem:** Application crashes immediately.
  - **Cause:** Missing or corrupt CSV files.
  - **Solution:** Ensure `students.csv`, `courses.csv` etc. are in the executable folder.

15.2 Data Operation Issues

- **Problem:** "Data not saved" error.
  - **Cause:** Permission denied or file locked.
  - **Solution:** Check permissions. Close CSV files if open in Excel.
- **Problem:** Corrupted characters.
  - **Cause:** Encoding mismatch.
  - **Solution:** Run `chcp 65001` on Windows for UTF-8 support.

15.3 Logic & Usage Issues

- **Problem:** "Capacity full" error.
  - **Cause:** Course reached max students.
  - **Solution:** Increase limit via *Set Course Capacity* or choose another course.
- **Problem:** "Prerequisites not met".

- **Cause:** Missing required prior courses.
- **Solution:** Complete prerequisites first. Check *Curriculum Management*.

- **Problem:** Cannot delete Student/Professor.
  - **Cause:** Active records exist (Data Integrity).
  - **Solution:** Drop student enrollments or reassign professor courses first.

15.4 Display Issues

- **Problem:** Misaligned tables.
  - **Cause:** Narrow terminal or wrong font.
  - **Solution:** Expand window or use Monospace font (e.g., Consolas).

16 Appendix

16.1 Keyboard Shortcuts Summary

Context	Key	Action
Any menu	`0`	Go back / Exit
Any input	`Enter`	Confirm input
Confirmation	`y`	Yes
Confirmation	`n`	No

16.2 GPA Calculation Formula

$$GPA = \frac{\sum_{i=1}^{n}(Credit_i \times GradePoint_i)}{\sum_{i=1}^{n} Credit_i} \tag{1}$$

Where:

- $Credit_i$ = Credits for course i
- $GradePoint_i$ = Grade point value (4.0 for A, 3.7 for A-, etc.)
- n = Total number of courses

16.3 Semester Format Reference

Season Code	Time Period
SPRING	January - May
SUMMER	June - August
FALL	September - December

Example: `2024-FALL`, `2025-SPRING`